Tomorrow's TODOs — 2026-06-29

Full Drug Database Download + Yajie One-Shot Full Screening

Database List (Full, not limited to HIV)

#	Database	Content	Size	Priority
1	**ChEMBL**	Full bioactivity data (all targets, all compounds)	~50GB	
2	**DrugBank**	All approved + experimental drugs with targets	~1GB	
3	**PubChem BioAssay**	All assay data	~50GB	
4	**BindingDB**	Full protein-ligand binding affinities	~2GB	
5	**PDBbind**	Protein-ligand complexes + binding affinities (most precise)	~5GB	

#	Database	Content	Size	Priority
6	**TTD** (Therapeutic Target Database)	Known therapeutic targets + corresponding drugs	~500MB	
7	**DrugCentral**	FDA-approved drug targets + indications	~1GB	
8	**Open Targets**	Target-disease associations (GWAS+literature)	~10GB	
9	**PharmGKB**	Pharmacogenomics (gene × drug × response)	~2GB	
10	**Stanford HIVDB**	HIV resistance mutations (retained)	<100MB	
11	**SIDER**	Drug side effect database	~1GB	
12	**STITCH**	Compound-protein interaction network	~20GB	

Disk Requirements

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Priority (must-have): ~105GB
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 Strongly recommended:  ~20GB
 Optional:              ~20GB
Buffer + decompression:  ~50GB

Total:                   ~200GB
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Preprocessing Pipeline

- ☒ All SMILES → standardization → ECFP4/MACCS fingerprints (screen_all_databases.py Phase 2)
- ☒ Unified schema: drug_id, target_id, assay_type, value, units, source_db
- ☒ Deduplication (same drug-target pair appearing in multiple databases = expert consensus)

Yajie Execution

- ☒ N experts (one per database + physicochemical properties + structural similarity)
- ☒ One-shot screening: all drugs × all targets (screen_all_databases.py)
- ☒ Output:
  - High-consensus drug-target pairs (cross-database validation) → mt_high_confidence.csv
  - Database quality report (which database has most contradictions) → mt_quality_report.json
  - Emerging targets (high novelty, low consensus → Spring revival candidates) → mt_novel_candidates.csv

Validation

- ☐ Are known HIV drugs scored high by Yajie?
- ☐ Does Yajie discover known cross-domain targets (HIV/autoimmune/inflammation shared targets)?
- ☐ False Positive analysis (high score but no literature support → new discovery candidates)

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arXiv Submission

- ☐ Paper 1 (Yajie) — PDF compiled (including S1-S8)
- ☐ Paper 2 (Spring) — PDF compiled (13 pages)

Completed

- ☒ Spring code (1551 lines) + Yajie fit() + HIV audit prototype
- ☒ 9 paper frameworks + 18-layer game theory + 445 tests all passing
- ☒ **Full Drug Database Download Script** `drug-module/scripts/download_databases.py` (~1050 lines)
  - 12 databases full coverage (ChEMBL, DrugBank, PubChem BioAssay, BindingDB, TTD, Stanford HIVDB, PDBbind, DrugCentral, Open Targets, PharmGKB, SIDER, STITCH)
  - Tiered download (Tier 1/2/3) approx. 205GB
  - provenance.json written per database (SHA256 checksum, download date, version, URL)

  - Progress bar + ETA, resume support, authentication database placeholders
  - Auto-parsing: ChEMBL SQLite→Parquet, DrugBank XML→Parquet, BindingDB TSV→Parquet, PubChem PUG REST API, Open Targets GraphQL API
- ☒ **Yajie Full Database Screening Script** `drug-module/scripts/screen_all_databases.p` (~930 lines)
  - Phase 1: Load all database Parquet/CSV + provenance metadata
  - Phase 2: SMILES→ECFP4 fingerprint standardization + target→UniProt ID mapping
  - Phase 3: Drug-target unified evidence matrix (rows = drug-target pairs, columns = database experts)
  - Phase 4: Yajie multi-expert consensus scoring (Theorem 1: CLEAN/NOISY/AMBIGUOUS classification)
  - Phase 5: 8 output files (MT Gold Standard CSV, classification summary, database consistency matrix, high-confidence pairs, disagreement report, novel candidates, quality report JSON, provenance audit JSON)
  - Fully traceable: each pair records source_databases, database_links, download_date, mt_report_version